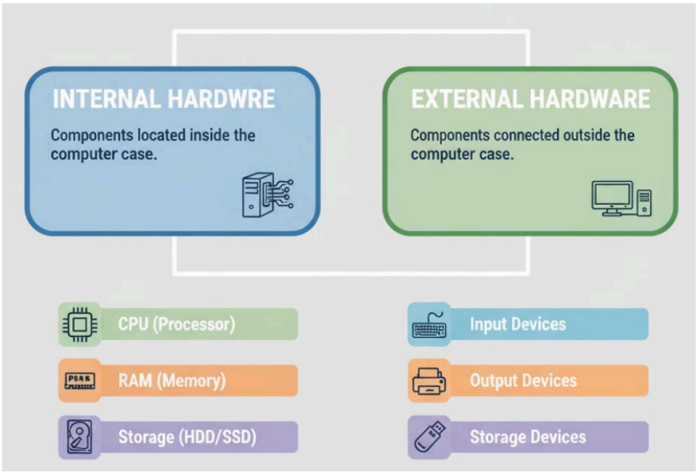
SWE 4101
Hardware Components & Peripherals

Lec - 2

What is Computer Hardware



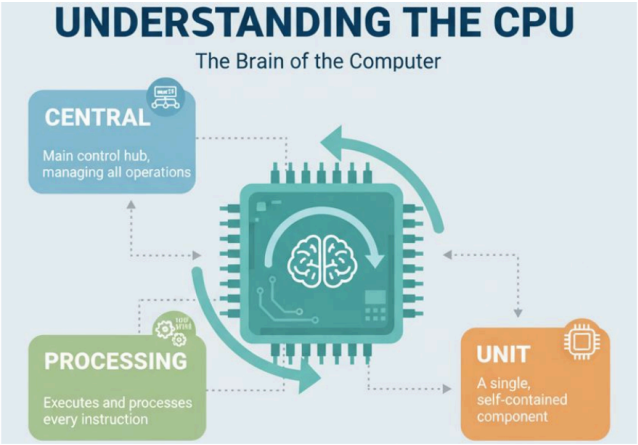
✔ Hardware is the **physical**, **tangible** parts of a computer system that you can **touch.**



CPU (Central Processing Unit)



✔ **computer's brain**. The CPU was invented by Intel in 1971.
✔ handles all the tasks you tell the computer to do, like opening a game or writing a document.



CPU (Central Processing Unit)

✔ **Types of CPU**

Based on purpose

General Purpose CPU : A CPU designed to handle a wide variety of tasks and applications, not limited to a specific function. For example Intel Core i7, AMD Ryzen 7.

Special Purpose CPU : A CPU designed for a specific task or application, optimized for that function rather than general use. GPU processors in graphics cards (like NVIDIA RTX) for rendering graphics, or microcontrollers in washing machines.

Based on Architecture

CISC (Complex Instruction Set Computer): A CPU architecture with a large set of instructions, allowing a single instruction to execute multiple low-level operations. Intel x86 processors — one instruction can move, add, and store data in a single command

RISC (Reduced Instruction Set Computer): A CPU architecture with a small set of simple instructions that execute very quickly, often requiring multiple instructions for complex tasks. ARM processors in smartphones (like Apple M1, Qualcomm Snapdragon).

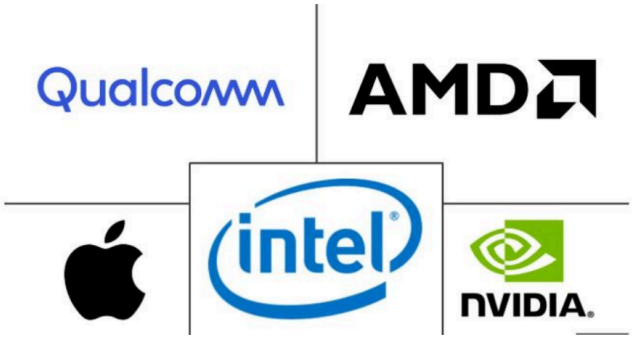
Based on Core

Single-Core CPU: A CPU with only one processing unit (core) that can execute one instruction at a time. Early Intel Pentium 4 processors.

Multi-Core CPU: A CPU with two or more cores, allowing it to execute multiple instructions simultaneously, improving multitasking and performance. Intel Core i9 (8-core/16-thread), AMD Ryzen 9 (12-core/24-thread).

CPU (Central Processing Unit)

✔ **Major CPU Manufacturers**



Memory Unit

✔ One of the most vital components of a computer system.
✔ Without memory, a computer would not be able to perform even the simplest tasks.

Understanding Memory Units

✔ The storage capacity of computer memory is measured in **bytes**.
✔ Each byte consists of **8 bits**, and each bit represents a binary value—either 0 or 1.
✔ The standard memory units are as follows:

Unit	Symbol	Equivalent
Bit	b	Smallest unit of data (0 or 1)
Byte	B	8 bits
Kilobyte	KB	1,024 bytes
Megabyte	MB	1,024 KB
Gigabyte	GB	1,024 MB
Terabyte	TB	1,024 GB
Petabyte	PB	1,024 TB
Exabyte	EB	1,024 PB

Memory Unit

Primary Memory (RAM)

How Much RAM Do You Need?

- ✔ The amount of RAM you need depends on your usage.
- ✔ Having more RAM makes your computer run faster and more smoothly.



You have multiple browser tabs, a video editing software, and a music player running at the same time. Suddenly, your computer freezes and takes a long time to respond.
Why did this happen?

Memory Unit

Primary Memory (ROM)

- ✔ Read-only memory (ROM), is a type of **non-volatile memory** in a computer, which means the data stored in it **remains even when the power is turned off**.
- ✔ primarily used to store **permanent instructions** that the computer needs to start and operate, such as the BIOS or firmware.

PROM (Programmable ROM)

- ✔ A type of ROM that can be programmed **only once** after manufacturing.
  - Once written, the data **cannot be changed**

Example: Used in electronic devices where permanent instructions are needed, like calculators or microwave controllers.

EPROM (Erasable Programmable ROM)

- ✔ A type of ROM that can be **erased and reprogrammed**.
  - Data is erased using **ultraviolet (UV) light**
  - Can be reprogrammed multiple times

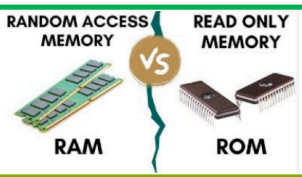
Example: Used in BIOS chips of computers in earlier days

EEPROM (Electrically Erasable Programmable ROM)

- ✔ A type of ROM that can be **erased and reprogrammed electrically** without removing it from the system.
  - Data is erased using **electrical signals**
  - Can be rewritten multiple times

Example: Modern BIOS chips, memory cards, and embedded systems

RAM vs ROM



Feature	RAM (Random Access Memory)	ROM (Read-Only Memory)
Type of Memory	Volatile	Non-volatile
Data Loss	Data is lost when power is off	Data remains safe
Use	Temporary storage	Permanent instructions
Speed	Very fast	Slower than RAM
Editable	Can be changed	Cannot be easily changed
Storage Capacity	High	Low
Cost	Expensive	Cheaper
Working Stage	Active during processing	Active during booting
Example	DDR4, DDR5	BIOS, Firmware

Memory Unit

Secondary Memory (Magnetic Tape)

- ✔ A magnetic tape stores data on a long plastic strip.
- ✔ It looks like the tape inside a cassette.
- ✔ Can save a lot of data for a long time. slow but cheap.
- ✔ Industries use it to keep backup files.

Key Features of Magnetic Tape:

- Stores very large data (multiple terabytes)
- Non-volatile; retains data without power
- Sequential data storage (one after another)
- Slow read/write speed
- Low-cost for long-term storage
- Portable and easy to carry
- Requires a tape drive to read/write
- Type of magnetic storage device



Memory Unit

Secondary Memory (Others)



Floppy Disk:
- ✔ Stores 1.44 MB, magnetic storage
- ✔ Lightweight and portable
- ✔ Slow read/write, requires floppy drive
- ✔ Obsolete; replaced by better devices



Compact Disc (CD):
- ✔ Stores up to 700 MB, non-volatile
- ✔ Uses laser (optical) storage
- ✔ Lightweight and portable
- ✔ Requires CD drive
- ✔ Low-cost storage option



SD Card:
- ✔ Stores data from a few MB to 1 TB+
- ✔ Non-volatile; retains data without power
- ✔ No moving parts; silent operation
- ✔ Small and lightweight
- ✔ Used in phones, cameras, tablets
- ✔ Removable and reusable
- ✔ Faster read/write than CDs/DVDs



DVD:
- ✔ Stores 4.7–8.5 GB, non-volatile
- ✔ Optical storage for videos, software, files
- ✔ Lightweight and portable
- ✔ Requires DVD drive
- ✔ Rewritable DVDs (DVD-RW) are reusable and cheap



USB Flash Drive:
- ✔ Stores data from 2 GB to 1 TB+
- ✔ No moving parts; durable
- ✔ Uses solid-state storage
- ✔ Fast data transfer
- ✔ Reusable; can delete and save data multiple times
- ✔ Connects via USB



Blu-ray Disc:
- ✔ Stores 25 GB (single layer) or 50 GB (dual layer)
- ✔ Non-volatile; retains data without power
- ✔ Uses blue laser for reading/writing
- ✔ Better picture and sound than DVD
- ✔ Lightweight, portable, and scratch-resistant
- ✔ Works only in Blu-ray players or drives

Memory Unit

Cache Memory

- ✔ Cache is the temporary memory officially termed “CPU cache memory.”
- ✔ is a high-speed memory location built into or very close to a computer’s processor.
- ✔ stores copies of frequently used data and instructions so that the CPU can access them much more quickly than if it had to fetch them from main memory (RAM) or from disk.

Why Cache Matters (Performance Impact)

- ✔ When the CPU needs to access data, it checks the cache first.
- ✔ If the data is found (“cache hit”), the access is extremely fast.
- ✔ If it’s not found (“cache miss”), the system must fetch it from slower memory or storage which delays performance.

Levels of Cache

Modern CPUs typically implement multiple cache levels to balance speed and size:

- ✔ **L1 Cache**: The smallest and fastest—built directly into the CPU. It holds the data most likely to be used next.
- ✔ **L2 Cache**: A larger but slightly slower cache that the CPU checks if data isn’t found in L1.
- ✔ **L3 and beyond / Other caches**: Some systems include L3 or even L4. There is also disk cache (on hard drives/SSDs) and GPU cache in graphics systems.

When you’re buying or evaluating a computer, cache is one of many specs—but not always the most visible or advertised.

- ✔ **Cache size**: A larger cache can hold more data, but may also have higher latency (slower access).
- ✔ **Latency/speed**: The access speed of the cache matters more than size.